SOP-33: Sibling and Family Arbitration Operations

Version 2.0 · 2026-08-08 · Owner: PPC Lead · Supersedes SOP-33 v1.0 (2026-07-24) in full · Built session 16, Wave 5A · Thresholds, bands, scenarios and verdicts live in the Master PPC Decision Criteria System v4.0 only; composite weights and factor definitions live in the Account Rollup Workbook v1.0 tab A1 operating under v4.0; this SOP carries zero local thresholds

1. Verdict and Scope

VERDICT: when two family products can serve the same root, ownership is decided by one arithmetic product read off one instrument, and until that number is on the record the guards hold on both siblings at once. The composite is **margin dollars per order multiplied by root CVR multiplied by inventory depth in days**, computed on the Account Rollup Workbook v1.0 tab A1, and its output is a relative index, never a per-click dollar value. Version 2.0 corrects the v1.0 composite method, which computed a two-term "value per click" and then applied a separate inventory factor; that method does not reproduce the instrument's own live cell and is retired. Version 2.0 also retires five v1.0 verdict strings that belong to no closed vocabulary, expands 7 procedures to 14, 6 failure modes to 12, 1 worked example to 4, and a 5-row verdict mapping to the full 25-string closed vocabulary with the two v4.0-only cross-map rows the corpus requires. This SOP governs the whole arbitration life of a contested root: weekly detection, finding intake, guard enforcement, evidence assembly, the dated probe when a sibling has no served evidence, the composite computation, the mandatory structural inspection that runs before any auction explanation is accepted, the declaration and its same-day walls, the non-owner posture, twin-set coordination, re-arbitration on named drivers, temporary stewardship on an owner stockout, and the quarterly drift review.

1.1 Why the composite is three terms and not two

Margin dollars set what a sale is worth. Root CVR sets how often a click becomes a sale. Inventory depth sets how long the product can hold the pressure that ownership implies. A sibling that wins on the first two and cannot supply the third is an owner in name only, because the root it owns will be re-contested inside a month by its own stockout. Multiplying all three in one expression is what makes depth a cost of ownership rather than a caveat attached to it. The instrument's own live cell proves the form: the Account Rollup Workbook v1.0 tab A1 reads 58.3358125 for the Bamboo 4-piece row, and 28.37 x 0.04375 x 47 = 58.3358125 to seven decimal places, which no two-term-plus-factor arrangement reproduces. PPC Strategic Doctrine v1.0 Section 5.2 case 4 states the same three-term product in words: one family member owns the root on a margin x CVR x inventory depth composite and the sibling defends at floor.

1.2 The three states of a shared root, and what each one costs

State	Definition	What it costs while it persists
UNDECLARED	A root appears in two or more siblings' campaigns or served terms and no declaration record exists.	Both siblings bid the same auction with the same money. Neither may open or lift a rank push. Existing spend continues at current levels, so the cost is opportunity plus auction self-competition, not an immediate cut. Master PPC Decision Criteria System v4.0 Section 5 gate G11 makes this a portfolio decision, not two product decisions.
DECLARED	One sibling holds the root on a recorded composite table, with a dated record and named re-arbitration triggers.	Nothing, while the drivers hold. The non-owner keeps its own modifier space and defends at floor there. The cost of a declaration is the walls it requires and the review it obliges.
STEWARDED	The declared owner is out of stock; a dated, reverting grant places the root with the strongest sibling until the owner restocks and recovers.	A live reversion obligation. Stewardship that outlives its date converts an inventory event into a silent ownership change, which is the F5 condition and the reason every grant carries a computed reversion date at the moment it is issued.

1.3 What this SOP does not decide, named by component

Adjacent component	The seam, stated so it is owned
Criteria System v4.0 (#8)	Decides every threshold, band, gate, scenario and verdict string. This SOP computes and declares; it never sets a number. A threshold appearing in this SOP instead of v4.0 is a defect.
Account Rollup Workbook v1.0 tab A1	Owns the composite formula, its factor definitions, the grid layout and the S-series finding format. This SOP operates the instrument and returns its inputs and its declaration; it does not redefine the arithmetic. Per the Cross-Reference Ledger row 33, composite weights are the instrument's.
#29 Negation architecture	Owns wall doctrine, wall levels, the reverse test, removal control and the monthly wall audit. This SOP issues the declaration that causes a wall and consumes the confirmation line back; per the Ledger row 29 the two contracts are that SOP's I-8 inbound and I-9 outbound. Cite by Ledger row, not by procedure number: the v2.5 correction register records the sibling-wall move from P5 to P7 in v2.0, and the Ledger row 33 still carries the superseded number.
#23 Ranking-push operations	Owns push sizing, funding, standup, in-flight operation and stop-loss. The guard in this SOP is a precondition on that SOP, recorded as its I-3 in the Ledger row 23, whose non-arrival consequence is an indefinite hold on both siblings. This SOP never sizes, funds or stops a push.
#12 Exact campaign operations	Runs the G11 declared-owner check before any scale action on a shared term, per the Ledger row 33 downstream cell. This SOP supplies the declared owner; it issues no campaign-level bid, budget or status edit.
#24 Re-ranking and recovery	Owns baseline restoration and the post-restock recovery play. This SOP reads the restock date and the recovery class to compute a stewardship reversion date; it does not run the recovery.
#32 Deal and event window	Owns the event plan, arming and reconciliation. This SOP requires a contested root to be named in an event plan touching either sibling, so that a window cannot adjudicate ownership by accident. It does not write event plans.
#22 Launch-stage PPC	Owns the launch handoff. This SOP receives the newcomer's keyword set and scans it against declared roots; it does not govern any part of the launch sequence.
Envelope and funding	Cross-product push funding is ranked on the Account Rollup Workbook v1.0 tab A2 and the product's T0 zone H queue. Per Ledger conflict C3, component #31 carries no numbered procedures, so this SOP names the instrument and tab and never a #31 procedure number.
#38 Bulk mechanics	Owns staging format, upload and T+1 verification. Every wall and every campaign-level consequence of a declaration deploys through that contract; nothing in this SOP is edited in a console.

1.4 Depth statement

This component fills the SOP Build Kit v1.0 range without padding. The reason is that arbitration is a low-frequency, high-consequence decision with a long tail of states: a declaration is issued perhaps a dozen times a year across five brands, but each one binds two products' plans under v4.0 Section 12 precedence rule P9, survives until a named driver moves it, and has three failure directions that all look like success from inside one product's workbook. The procedures below therefore carry the evidence-qualification detail that a once-a-quarter task cannot rely on memory for.

2. Trigger Conditions

Every trigger names the artifact that detects it and the maximum lag from signal to procedure start. Lag is counted from the moment the artifact is readable, not from the moment someone opens it.

Trigger	Detecting artifact and signal	Maximum lag	Procedure
Shared-root finding	The cross-product registry scan across product Command Workbook tab T2 block 8 columns and the search-term join: any root live in two or more siblings' campaigns, or converting in two or more siblings' served terms.	Same cascade. The scan runs inside the Monday pass after the product workbooks refresh at 09:00.	P1 detection, then P2 intake and guard arming in the same session
Family Conflict cell opens	Command Workbook tab T2 block 2 column "Family Conflict" moves off its clean state on any oversight row, for example the live B4 reading "OPEN: shared with B6 sibling, owner undeclared".	Same day. The cell is a gate column, not a metric column.	P2 intake; P3 guards armed before any verdict issues on that row
New sibling launch	The launch handoff from #22 lands a product into a family that already holds declared roots, with its keyword set attached.	Within one cascade of the handoff arriving.	P1 scan restricted to the newcomer's keyword set against the declaration register
Composite driver moves	Any of the four inputs changes: a price or cost move on the D1 margin table, a root CVR read crossing its recorded re-arbitration trigger, an inventory band change on tab T2 block 2, or a variation retirement on the D2 map.	Same cascade for margin and CVR. Same day for an inventory band flip, because depth is a multiplier and a band flip moves the composite immediately.	P12 re-arbitration
Owner stockout	The declared owner's serving variation reaches v4.0 Section 5 gate G6 RED, or stockout is confirmed on the Economics and Context Pack D1 row and the D2 backup chain is exhausted.	Same day. The re-point runs first and only an exhausted backup chain reaches this SOP.	P13 stewardship grant with its reversion date computed at grant
Sibling CPC divergence	The A1 family grid divergence flag on a shared root, read against the v4.0 Section 4 rule X6 trigger.	Same cascade.	P7 structural inspection; a divergence on an undeclared root also opens P2
Declaration deadline	An open finding reaches the deadline recorded on it at intake with no declaration filed.	Same day the deadline passes.	P14 escalation path; guards stay armed, they do not lapse with the deadline
Quarterly review	The review date on the declaration register.	Within the review week.	P14 drift check across every live declaration

3. Preconditions and Gates

Each precondition names its manifest field group from the Master PPC Data Architecture v1.0 tab T7 and states the observable that proves it passed. A missing group suppresses its dependent step and the finding records what it could not see; it never degrades silently, per v4.0 Section 5 gate G8.

Manifest group	Requirement before the dependent step runs	Observable that proves it passed
B5 Coverage counts	Registry counts per match type current on both siblings, or the detection scan cannot distinguish a shared root from a reporting artifact.	Tab T2 block 8 exact, phrase, broad, auto and SB/SBV/SD/PAT counts populated on the contested root for both products, with the refresh timestamp inside the current cycle.
B8 Row economics	Margin dollars per order readable for the variation the root actually lands on, not the product blend.	The Economics and Context Pack v1.0 tab D1 carries a Contribution $/unit and a Version stamp on the serving variation row. If only the blended row carries a value, v4.0 Section 5 gate G9 rides PROVISIONAL onto the composite and onto the declaration record.
D2 Variation map	The root mapped to its targeted variation and its ranked backup chain on both siblings.	Tab D2 "Keyword-Variation Map" shows a Preferred SKU and a Backup SKU on the root's row with a status other than the unmapped state. Ledger Section 2 dependency D2 is OPEN across the corpus, so most rows read as pending and the depth input is capped to a product-level read, which the finding states.
A3 Pacing and A2 deployment	Current bid, budget and placement multiplier state on every campaign serving the root on both siblings.	Tab T3 columns for state, daily budget, bid and placement multipliers populated for each campaign the registry lists against the root. Absent state makes the P7 inspection unreadable, and an inspection that cannot read settings cannot exclude settings.
B3 Rank pack	Organic rank and target rank on the root for both siblings.	Tab T2 block 4 organic rank and target rank present. Rank does not enter the composite; it enters the record, because a declaration that hands a root to the weaker rank position needs that visible on its face.
Inventory feed	Days to stockout on the preferred SKU for both siblings, refreshed daily.	Tab T2 block 2 "Days to Stockout" carries a figure and the SKU it belongs to, for example the live B4 reading of 47 days on the KING-WHITE preferred SKU.

Gates in force, named by v4.0 number with the corpus alias at first use

Gate	Effect on this SOP
G11 Sibling conflict	v4.0 Section 5 "Layer 0: Gates (G1-G12). Nothing Proceeds Until These Pass" makes two siblings carrying active push objectives on the same term a portfolio decision, and holds both from scaling until the portfolio call assigns a lead and a support-or-stand-down posture. This gate is the reason this SOP exists, and the Family Conflict cell on tab T2 is where it is read.
G6 Inventory adequacy	Supplies the depth term. A RED band on the declared owner's serving variation routes to P13 after the re-point is exhausted. A YELLOW band does not by itself re-open a declaration; it moves the composite and is read at the next scheduled review unless it crosses a recorded re-arbitration trigger.

G9 Price stability	A price change on either sibling's serving variation inside the trailing window marks CVR reads PROVISIONAL. A composite computed on a provisional CVR is legal, but the flag is carried onto the declaration record in the same words the engine computed, per v4.0 Section 13.3 rule J7. Suppressing it is the F11 condition.
G2 Sample sufficiency	Sets what counts as evidence for the root CVR input. Twin sets pool evidence across the set for sufficiency, which is why P11 twin-set registration precedes the CVR read on any root that carries a twin.
G8 Packet completeness	A degraded packet downgrades the composite to a flag. The finding stays open, the guards stay armed and no declaration issues on a degraded read.
G12 MKL hygiene	The corpus alias for this number is the event freeze; v4.0 assigns G12 to MKL hygiene and Ledger conflict C9 records the divergence. Both apply here. Under the v4.0 sense, a root not carrying a syntax tag and a relevancy score routes to taxonomy maintenance and is not arbitrated. Under the corpus sense, a live event window on either sibling requires the contested root to be named in the event plan.

4. Step-Level Procedures

Fourteen procedures. Each carries a time estimate for a single contested root on a two-sibling family. Every navigation reference names the actual workbook, tab, block and column, so the sequence executes from the page.

P1. Weekly cross-product shared-root detection scan *(20 minutes for a five-brand account; 35 minutes on the first run of a quarter)*

1. Open the Account Rollup Workbook v1.0 after the product cascades have closed. The refresh order matters: product workbooks close Monday, the rollup refreshes after them, and a scan run before the rollup refresh reads last week's family grid.
2. Build the candidate list from two sources, not one. Source A: for each product Command Workbook, tab T2 filtered to Row Type = KW, read block 8 columns "Sibling CPC View (D6)" and "Term Owner (Family)". Source B: the search-term join under each keyword's campaign child rows, which exposes roots a product serves without carrying as a keyword. A root that appears in Source B only is the more expensive class, because nobody has it on a keyword row to look at.
3. Normalise to roots, not strings. Strip size, colour and count modifiers using the syntax tags in tab T2 block 1 column "Syntax Tags", which carry the root|size|colour|attribute form. "bamboo sheets", "bamboo sheets queen" and "queen bamboo sheet set" are one root and three terms; the root is what is arbitrated and the modifiers are not.
4. Test each normalised root against the two conditions in turn. Condition 1: the root, or any term under it, is live in two or more siblings' campaigns per the block 8 registry counts. Condition 2: the root, or any term under it, converted in two or more siblings' served terms in the trailing window. Either condition alone opens a finding. Both together raise the intake priority but do not change the procedure.
5. Exclude on evidence only. A root drops out of the candidate list for exactly three reasons: it already carries a live declaration and no driver has moved, one of the two products is not a family sibling under the A1 family definition, or the second appearance is a reporting artifact proven by a registry read showing no live campaign and no served term. Any other reason is not an exclusion; it is a finding with a note.
6. Record the scan even when it is empty. Zero-finding weeks are written to the declaration register with the date and the count of roots tested. A scan with no artifact is indistinguishable from a scan that never ran, which is the F12 condition.

> **Instrument gap to state in every scan record.** Tab T2 block 8 column "Sibling CPC View (D6)" reads as a pending build across the corpus, so Source A is currently thin and Source B carries the detection load. This is not an operator failure and it is not a reason to shorten the scan; it is a named feed absence that the T7 manifest already records, and the scan states which of the two sources actually produced each finding.

P2. Finding intake, evidence snapshot and guard arming *(15 minutes per finding)*

1. Open an S-series finding on the A1 family grid in the instrument's own finding format. The format is the Rollup's and is not redefined here.
2. Attach both siblings' current state on the root at the moment of opening, from tab T2: current bid and suggested bid with the gap, TOS modifier percentage, spend on the root for the window, term clicks, term orders, organic rank against target rank, and the inventory band with days to stockout and the SKU it belongs to. This snapshot is what the composite is later checked against, and it is taken before any guard changes behaviour.
3. Write the declaration deadline onto the finding. No threshold in v4.0 sets this interval, so the deadline is taken from the instrument's own review cadence for the family grid and the absence is logged as amendment item AM-2 rather than filled with an invented number.
4. Set the Family Conflict cell on tab T2 block 2 for the root on both siblings' workbooks to the open state, naming the counterpart product and the finding. The live B4 form is the model: an open marker, the sibling named, and the gate reference.
5. Set "Term Owner (Family)" in tab T2 block 8 to the undeclared state on both siblings. An empty cell and an undeclared cell are different states, and only one of them is readable by the G11 check that #12 runs.
6. Arm the guards per P3 in the same session. A finding opened without its guards armed leaves the account in the state the finding was raised to stop.

P3. Guard enforcement while undeclared *(5 minutes to arm; thereafter enforced by the weekly pass)*

1. No sibling opens a rank push on the contested root, and no live push on it is lifted. This is enforced as a precondition inside #23, recorded as that component's I-3 in the Ledger row 23; this SOP supplies the guard state, that SOP holds the push.
2. Existing spend continues at current levels. The guard freezes escalation; it does not cut. Cutting on a finding would make the arbitration itself a spend event and would corrupt the CVR evidence the composite is about to read.
3. The contested root is excluded from the cross-product funding queue on Account Rollup Workbook tab A2 until the finding closes. A queued ask on a contested root is withdrawn, not deferred, so the queue's rank order stays readable.
4. Any event plan under #32 touching either sibling names the contested root explicitly. If the plan does not name it, the root is out of the window: windows amplify what is decided and do not decide it.
5. Structural changes on the root freeze on both siblings: no new exact instance, no consolidation, no promotion or demotion. Bid and budget movement inside existing structure continues to be governed by the normal weekly verdict path.
6. Guards lift on exactly one event, which is a filed declaration record with its wall confirmation attached. They do not lift on the deadline passing, on a stale finding, or on either product's preference.

P4. Composite input assembly and source qualification *(25 minutes per sibling; 40 minutes where the variation map is unmapped)*

Four inputs, one per sibling. Each has a required source and a disqualifying condition. An input that cannot be qualified is not estimated.

Input	Required source	Qualification test	If it fails
Margin $ per order	Economics and Context Pack v1.0 tab D1, the row for the variation the root actually lands on per the D2 map.	The row carries Contribution $/unit and a Version stamp, and the variation is the one the D2 map names, not the product blend.	Read the blended row, carry the G9 PROVISIONAL flag onto the composite and the declaration record, and log the variation row as owed. Using a blended figure without the flag is F11.
Root CVR	Tab T2 block 5 cross-type totals for the root's terms, orders divided by clicks over the cycle window.	The clicks are from the sibling's own serving on the root, and the sample clears the v4.0 Section 5 gate G2 bar for a CVR read. Twin-set members pool per G2.	Run P5. Never impute a CVR from a sibling, from a modifier term, or from the product average.
Inventory depth in days	Tab T2 block 2 "Days to Stockout" on the preferred SKU named in the D2 map.	The figure names its SKU and that SKU is the D2 preferred SKU for the root.	Use the product-level band read, state the substitution on the finding, and cap the result to a structural read. Ledger dependency D2 is OPEN and this is its operational cost.

Window	The A1 grid's cycle window, applied identically to both siblings.	Both siblings read the same window. A composite comparing a 30-day CVR against a 7-day CVR is not a comparison.	Re-pull to a common window. v4.0 states no composite recency window; the absence is amendment item AM-5.

1. Assemble all four inputs for both siblings before computing anything. Computing one side while the other is still being assembled is how a partial read becomes a declaration.
2. Write each input into the A1 grid row with its source cell reference beside it. The grid's evidence table is the declaration's evidence, and an input without a source is not evidence.
3. Record every flag that rides: G9 provisional margin, a thin sample, a substituted depth read, a pooled twin-set CVR. v4.0 Section 13.3 rule J7 requires flags in the same words the engine computed, and confidence theatre is a process violation.

P5. The dated probe for a sibling with no served evidence *(20 minutes to design and stage; 2 to 3 cycles to run)*

1. Confirm the condition: the sibling has never served the root, so no CVR exists on it. A sibling that served the root and converted poorly is not a probe case; it has evidence and the evidence is bad.
2. Choose between two legal paths and record which one and why. Path A: run a dated probe and hold the finding open. Path B: declare the served sibling now with a dated re-check written onto the declaration record. Path B is correct when the served sibling's composite clears the unserved sibling's best arithmetically possible composite; compute that ceiling explicitly using the unserved sibling's margin, its depth and the highest CVR any family member has recorded on the root.
3. Size the probe to the evidence bar, not to the verdict string. The closed vocabulary caps a deliberate test at 50 clicks, and a CVR verdict needs a larger sample under v4.0 Section 5 gate G2. The probe therefore runs in tranches: tranche 1 to the verdict string's cap, then a second tranche to the CVR sufficiency bar. Tranche 1 alone cannot produce the composite's CVR input, and treating it as if it could is the F7 condition. The missing single verdict string for a probe sized to the CVR bar is amendment item AM-7.
4. Set the probe bid from the sibling's own Expected CPC reference in tab T2 block 6, not from the counterpart's CPC. Where the per-term Expected CPC has not landed, the account reference in the T7 manifest applies and the substitution is stated on the finding.
5. Write the probe's stop conditions before it opens: the click cap per tranche, the date, and the single condition that ends it early, which is the sibling's own inventory band leaving GREEN. A probe into a thinning variation buys a CVR the product cannot then own.
6. Feed the probe result into P4 as a normal root CVR input, carrying the sample-state flag. Then run P6.

P6. The composite computation on the A1 grid *(10 minutes once P4 is complete)*

1. Enter the four inputs per sibling into the A1 family grid row: Margin $/Order, Term CVR, Inventory Depth in days, and the window.
2. The grid computes Composite = Margin $/Order x Term CVR x Inventory Depth. Do not overwrite the formula cell. Composite units are relative and carry no currency symbol; the instrument states this and the v1.0 rendering of the same quantity as a per-click dollar value was wrong.
3. Read Share of Composite for each sibling. Share is what the record carries, because a share states the margin of the decision and a raw index does not.
4. Do not declare yet. P7 runs before any declaration, without exception, and the order is the point: a composite computed on unexamined settings will confidently declare the wrong owner, which is exactly what WE-2 demonstrates.
5. Where the two shares are close, the composite has not decided. v4.0 sets no minimum separation and its stated tiebreaks in Section 8.11 scenario S-B2 and Section 10 rule M5 are margin-dollar class and inventory depth, and margin-dollar class and earning-potential class, which do not agree with each other and can both fail to separate. Record the near-tie, hold the guards, and log amendment item AM-1. A forced call on a dead heat is a coin toss wearing an evidence table.

P7. Structural inspection before any auction explanation *(30 minutes; 50 minutes where a duplicate is found)*

Mandatory on every arbitration, not only on ones where a divergence flag fired. v4.0 Section 4 rule X6 sets the order and the order is fixed: an auction explanation is accepted only after the five settings causes above it have been excluded on evidence.

```
Step 1  Objective tags and intentionality
        Read tab T2 block 7 "Objective" and tab T3 "Objective Tag" on every campaign
        serving the root on both siblings.
        UNTAGGED on either side --> the divergence has no stated reason yet.
        Route to objective assignment; the composite waits.

Step 2  Placement multipliers and effective TOS CPC
        Read tab T3 "Placement Mult TOS %" and tab T4 "Effective TOS CPC $".
        Effective TOS CPC = bid x (1 + TOS multiplier).
        Compare against the sibling's own max profitable CPC = margin x CVR.
        Above it --> the divergence is a multiplier, not a market.

Step 3  Bidding strategy
        Read tab T3 "Bidding Strategy" on both sides.
        Different strategies on the same term --> the two products are not
        bidding the same auction and the CPCs are not comparable.

Step 4  Duplicate and legacy campaigns still bidding
        Read tab T2 block 8 "Duplicate Flag" and the campaign child rows.
        A live legacy instance --> its spend and its orders are inside the
        pulled CVR and margin reads; both inputs are contaminated.

Step 5  Targeted variation differences
        Read tab T2 block 2 "Targeted Variation(s)" against the D2 map.
        Different variations --> different margin rows, and the composite
        input taken from the blend is wrong on at least one side.

Step 6  Genuine auction position difference
        Only reachable when steps 1 to 5 are clean. Record the conclusion
        with the evidence that excluded each prior step.
```

1. Work the six steps in order and write a one-line finding for each, including the clean ones. A step recorded as clean without a value beside it is not a record.
2. Any step that fires sends the composite back to P4 for a re-pull of the affected input. It does not adjust the composite in place. A corrected input changes the arithmetic, and the arithmetic is what the declaration rests on.
3. Where step 4 or step 5 fires, the repair belongs to #12 as a structural change and to #29 for any wall consequence. This SOP stages neither; it names the defect on the finding and re-runs the composite once the repair lands.

4. Write the inspection result onto the finding even when nothing fires. The absence of settings archaeology is itself the evidence that admits an auction explanation, and it is the clause that makes the declaration defensible three months later.

P8. The declaration decision and its record *(20 minutes)*

1. The declaration is human-mandatory. Per the PPC Authority Matrix v2.0 row "Sibling declarations (S-series)", the PPC Lead decides, the incoming operator executes, and the Brand Management owner reviews. The arithmetic recommends; it does not declare.
2. Confirm the four preconditions on the face of the record: P4 inputs all qualified or flagged, P6 computed with a separation the reviewer accepts, P7 inspection complete with all six steps recorded, and the packet not degraded under v4.0 Section 5 gate G8.
3. Write the declaration record with six fields and no fewer: the root, the declared owner, the full composite table with both siblings' four inputs and their sources, the date, the named re-arbitration triggers, and every flag that rode in. A record without the composite table is the F2 condition, and it is the failure the whole component exists to prevent.
4. Compute the re-arbitration triggers arithmetically, one per input. For the non-owner, solve the composite for each input in turn holding the other two fixed, giving the CVR, the depth and the margin at which the non-owner would equal the owner. These three figures are the triggers. A trigger stated as a direction rather than a number cannot fire.
5. Issue the wall instruction to #29 in the same session, per P9.
6. Update both siblings' tab T2 block 8 "Term Owner (Family)" to the declared owner and clear the Family Conflict cell in tab T2 block 2 on both, but only after the wall confirmation returns. Clearing the gate cell before the wall lands leaves #12 free to scale a root whose fences are not up.
7. Log the declaration in tab T6 with a dated prediction under v4.0 Section 14 rule V1: the metric, the direction, the magnitude and the horizon. The standard prediction for a declaration is the family blended CPC on the root and the family rank capture, because those are the two things a correct declaration moves and an incorrect one does not.

P9. Wall issue and confirmation *(10 minutes to issue; confirmation returns same day)*

1. Send #29 the declaration with the full source list: every campaign on the non-owner that can serve the root, across broad, phrase, auto and any sibling discovery structure, not only the ones currently serving it. A partial source list produces a partial fence and the term simply walks to the unfenced source.
2. The wall level, the wall mechanics and the reverse test are that component's, per the Ledger row 29. This SOP names the root and the sources and does not specify the negative match type.
3. The wall covers the root. It does not cover the non-owner's model-specific variants; see P10. An instruction that hands #29 the modifier space along with the root is what produces the F4 condition.
4. Require the confirmation line back the same day. Per the Ledger row 29, that return is that component's I-9. The declaration's gate artifact is its record with the wall confirmation attached, and until the confirmation is attached the declaration is drafted, not filed.
5. On a re-arbitration, the instruction is two-sided: the outgoing owner's discovery gains the wall and the incoming owner's loses it, both dated to the declaration change. The removal side routes through that component's removal control, which is a named case class there.

P10. The non-owner posture and the modifier-space boundary *(15 minutes)*

1. State the boundary in the declaration record in the same words every time: ownership is of the root, not of the modifier space that names the sibling. A 6-piece set does not lose its own count modifier because a 4-piece set owns the family root.
2. The non-owner holds exact defence at floor on its own model-specific variants, which are the root plus the size, count, colour or model terms that identify it. The posture verdict is DEFEND AT FLOOR, quoted from v4.0 Section 13.1 "The Closed Verdict Vocabulary", and the instrument states the same rule as the sibling defending at floor on the root.
3. Enumerate the protected variants on the record at declaration time, from tab T2 block 1 "Syntax Tags". Enumerating them later, after a wall has already suppressed them, is a restoration exercise rather than a boundary.
4. The floor itself is discovered and held by #28's ladder mechanics and executed by #12. This SOP names which terms sit at floor; it sets no bid.
5. Check the boundary at every wall audit cycle by reading the non-owner's impressions on its enumerated variants. A collapse on those terms after a wall build is the signal that the wall was built one level too wide, and it is the F4 detection.

P11. Twin-set registration and deployment sequencing *(15 minutes per family)*

1. Read tab T2 block 1 "Twin-Set ID" for every term under the contested root on both siblings. Twins are detected by the normalised-token signature, so plural and word-order variants of one term carry one identifier.
2. Register the twin-set identifiers across the family in the A1 twin-set registry so the two siblings' structures are visible as twinned rather than as unrelated campaigns that happen to look similar.
3. Pool evidence across the twin set for sufficiency before the P4 CVR read, per v4.0 Section 5 gate G2. A root that is thin on each twin separately can be sufficient across the set, and reading them separately manufactures a probe case that does not exist.
4. Enforce the sequencing rule: family-level experiments do not run simultaneously on twinned structures. Where one twin changes, flag the other for a read rather than changing both, because two simultaneous changes across a twin set produce one uninterpretable result.
5. Record the twin-set state on the declaration. A declaration on a root whose twins run different postures is a declaration that will be contradicted by its own family's structure inside two cycles.

P12. Re-arbitration on a named driver *(45 minutes; the composite and inspection re-run in full)*

1. Confirm the trigger against the declaration record's named numbers. A driver that moved but did not reach its recorded trigger is noted at the quarterly review and does not re-open the declaration.
2. Re-run P4 in full on current inputs for both siblings. Do not carry forward the prior composite's inputs, including the ones that did not move, because the window has changed and a mixed-window composite is not a comparison.
3. Re-run P7 in full. Settings drift between arbitrations at least as often as economics do, and a re-arbitration that skips the inspection inherits whatever archaeology accumulated since the last one.
4. Re-run P6 and compare both composites side by side on the record. The transfer log carries both tables, not just the new one, so the reason the owner changed is legible without opening two documents.
5. Where the owner changes, walls move the same day per P9 step 5 and push rights transfer with the declaration. Where the owner does not change, the declaration is re-dated with the refreshed table and new triggers, which is a real outcome and is recorded as one.
6. Update the tab T2 block 8 "Term Owner (Family)" cell on both siblings and log the transfer in tab T6 with a fresh prediction.

P13. Temporary stewardship on an owner stockout *(25 minutes at grant; 5 minutes per day while live)*

1. Confirm the re-point is exhausted first. Under v4.0 Section 9 rule L9, a thinning variation re-points ranking spend to the declared backup, and that move belongs to #12. Stewardship is only reached when the backup chain on the D2 map cannot carry the root.
2. Compute both siblings' composites at the moment of grant and record them. The owner's composite will have collapsed, because depth is a multiplier and a stockout drives it toward zero. State this on the record explicitly: the steward's advantage at grant is an inventory artifact, not an ownership case, and that sentence is what keeps the grant reverting.

3. Compute the reversion date at grant, not later. The reversion is the owner's confirmed restock plus its recovery period, and the recovery period is read from #24. No v4.0 threshold states a stewardship reversion interval independent of the post-restock recovery mode, so the interval is taken from the recovery contract and the gap is logged as amendment item AM-3.
4. Write the grant with five fields: root, steward, grant date, reversion date, and the owner's restock ETA that the reversion date was computed from. A grant missing the reversion date is the F5 condition at the moment it is written, not later.
5. Issue the walls per P9, dated to the grant and to the reversion, so the wall move at reversion is already staged rather than remembered.
6. Check the reversion date daily against the restock state while the grant is live. Where the restock ETA moves, the reversion date moves with it and the change is logged; the reversion is a computed date, not a fixed one.
7. At reversion, the root returns to the declared owner. If the steward's position has genuinely strengthened, that is a re-arbitration case under P12 and it runs after the reversion, not instead of it. Stewardship never converts to ownership by continuing.

P14. Quarterly declaration review, register maintenance and escalation *(90 minutes per quarter across the account)*

1. Read every live declaration against its recorded triggers using current inputs. This is a driver check, not a re-arbitration: the output is a list of declarations whose drivers moved past a trigger without a P12 event having fired.
2. Route each of those to P12. The quarterly review exists to catch what the triggers missed, and a review that finds drift and does not route it has converted a control into a report.
3. Re-date declarations whose drivers held, with the refreshed composite table attached. A declaration carrying a stale table is indistinguishable at a glance from one carrying a current one.
4. Reconcile the register against the instrument: every root the A1 grid shows as contested has a finding, every finding has a state, and every declared root reads the same owner in the A1 grid and in both siblings' tab T2 block 8 cells. A disagreement between the rollup and a product workbook is resolved in favour of the declaration record, and the divergence is logged.
5. Produce the review artifact as a register diff: declarations added, changed, re-dated and unchanged, with the composite share for each.
6. Escalate on two conditions. First, any finding open past its deadline, escalated with the days-open counted, because every undeclared week is the named failure pattern running live. Second, any declaration that has been re-arbitrated twice inside one review period, because a root that will not settle is a signal about the family's structure rather than about the arithmetic. No v4.0 threshold sets an escalation clock for the first condition; the gap is amendment item AM-12.
7. Route validated amendment items from the quarter to the threshold owner per the Authority Matrix v2.0 row "v4.0 threshold changes", where the CEO decides and the PPC Lead executes and reviews.

> **Cadence summary.** P1 weekly with the Monday cascade. P2, P3, P4, P6, P7 on finding, same cascade. P5 across 2 to 3 cycles. P8, P9, P10, P11 on declaration, same session and same day for walls. P12 on a named driver. P13 same day on an exhausted backup chain, then daily while live. P14 quarterly.

5. Decision Points: The Full Verdict-to-Execution Mapping

All verdict issuance, thresholds and scenario logic live in v4.0. This table maps the complete twenty-five-string closed vocabulary of the Keyword Decision Record Template v3.0 "Bands & Verdicts Reference" sheet, which governs per Ledger conflict C10 and matches exemplar SOP-12 v2.0 Section 5, to what this SOP does when each string lands on a row carrying a contested or declared root. Where a string diverges from the v4.0 Section 13.1 "The Closed Verdict Vocabulary" wording, the v4.0 string is carried in its own column so neither citation is dropped. Where execution belongs to an adjacent component, the boundary is stated so no verdict falls into a seam.

Verdict (KDR Template v3.0, verbatim)	v4.0 Section 13.1 cross-map	Arbitration execution
QUARANTINE: DATA REFRESH	FLAG: DATA REPAIR	No composite runs on a quarantined row. The finding stays open, the P3 guards stay armed, and the input carries forward to the next clean cycle. The refresh task belongs to Data Operations; this SOP records which composite input it blocked.
UNLOCK TRAFFIC	Same string absent; nearest v4.0 member is EXTEND TEST at the sufficiency branch	No arbitration execution. On a contested root, a bid step to unlock traffic is escalation and is held by P3 step 1. On a declared root, execution is #12's on the owner's campaign only.
FIX BUDGET (TRUNCATED)	No v4.0 member; the condition is gate G3	No arbitration execution and no guard block: budget assurance is a gate lever, not escalation. The finding records the truncated period, because the root CVR read for P4 is void across it.
CONTINUE TEST TO 50 CLICKS AT CAPPED BID	FUND DELIBERATE TEST (50-CLICK CAP)	The P5 probe tranche-1 verdict for an unserved sibling. This SOP designs the probe, sets its bid reference, its cap and its stop conditions; the campaign-level execution is #12's. Tranche 1 alone cannot produce the composite's CVR input.
FUND TEST (UNFUNDED PRIMARY)	FUND DELIBERATE TEST (50-CLICK CAP)	The P5 opening verdict where the unserved sibling has never funded the root at all. Same boundary as the row above.
RANK PUSH: FUND SIZED PUSH	RANK PUSH AT CONTROLLED CPC	Blocked on a contested root by P3 step 1, which is a precondition inside #23 recorded as its I-3. Released by a filed declaration, and released only to the declared owner. Sizing, funding and standup are #23's; this SOP supplies the guard state and the owner.
RECOVERY PUSH	RANK PUSH AT CONTROLLED CPC at scenario S-R13	Same block and same release. A recovery push on a root under stewardship belongs to the declared owner after reversion, never to the steward, because a recovery push by a steward is how stewardship converts silently.
TRAFFIC DEFICIT: FEED TO REQUIRED CLICKS	No separate v4.0 member; the condition is rule X4 and scenario S-R3	Held while undeclared as a push lift under P3 step 1. On a declared root it runs on the owner only. A non-owner is not fed to required clicks on a root it does not own; that is the same auction paid for twice.
FIX CONVERSION FIRST; DEFER PUSH	FIX CONVERSION FIRST; DEFER PUSH	No arbitration execution. Recorded on the finding, because a conversion deficit on the root moves the CVR input directly and is therefore written as a named re-arbitration driver at P8 step 4.
PLACEMENT FIX FIRST	No v4.0 member; the condition is rule L3 and scenario S-R7	Frequently the correct output of P7 step 2. This SOP names the defect on the finding and returns the composite to P4 once the fix lands. The placement program is #28's and the campaign rows are #12's.
RE-POINT VARIATION	RE-POINT VARIATION	Runs before stewardship is reachable, per P13 step 1. This SOP consumes the outcome: a successful re-point holds ownership and moves the depth input; an exhausted backup chain is what reaches P13.
TAPER: RANK ACHIEVED, STEP BIDS DOWN	TAPER TO FLOOR (POST-OBJECTIVE)	No arbitration execution. A taper on the owner's root moves neither ownership nor the composite, since none of the three composite inputs is a bid. Execution is #12's against #28's ladder.
HOLD; DEFEND LIGHTLY AT RANK	No v4.0 member; the posture sits inside DEFEND AT FLOOR and scenario S-D1	No arbitration execution. Recorded as the owner's post-objective posture where it applies, so the quarterly review reads the owner's actual state rather than its declared intent.
DEFEND AT FLOOR	DEFEND AT FLOOR	The non-owner posture verdict issued at P10 on the enumerated model-specific variants. This SOP names which terms sit at floor and enumerates them on the declaration record; the floor value is #28's discovery and #12's execution.
BID DOWN TO PROFITABLE CPC; CAP SYNTAX	BID DOWN TO PROFITABLE CPC; CAP SYNTAX	No arbitration execution. Where P7 step 2 finds an effective top-of-search CPC above the sibling's own ceiling, the repair routes here and the composite waits for the corrected read. Ownership is not a licence: a declared owner running above its ceiling on its own root still carries this verdict.
INCREMENTALITY LADDER	No v4.0 member; the mechanism is rule X12	No arbitration execution. Owned by #35. A live incrementality test on a contested root suspends the P4 CVR read for its duration, because the test is deliberately moving the number the composite reads.
ADD EXACT + RESTRUCTURE	ADD EXACT + RESTRUCTURE	No arbitration execution. Frequently the repair that P7 step 4 or step 5 produces. Owned by #12; the composite re-runs after it lands and a validated window accrues.
CONSOLIDATE DUPLICATES	ADD EXACT + RESTRUCTURE at scenario S-B4	No arbitration execution. The P7 step 4 repair. Consolidation inside one sibling is not arbitration; arbitration is between products. Owned by #12, whose survivor rule and precedence order govern.
GRADUATE FROM DISCOVERY	GRADUATE + STEER (Ledger conflict C10)	No arbitration execution, with one interlock: a graduation standing up an exact on a root another sibling owns is held until the declaration is read, otherwise the harvest pipeline re-creates the contest the declaration just closed. Owned by #30.
NEGATE (PRE-LOAD / REACTIVE / STEERING)	CUT / NEGATE (Ledger conflict C10)	The wall class this SOP causes at P9 and moves at P12. Doctrine, wall level, the reverse test and removal control are #29's per the Ledger row 29. This SOP names the root and the full source list and specifies no negative match type.
ENTER SBV ARBITRAGE	FORMAT SUBSTITUTION (SBV/SD)	No arbitration execution. A format substitution does not resolve ownership: both siblings still contest the same root on the cheaper surface. Owned by #17-19 with the paired step-down at #12.
OPEN / ROTATE / EXIT CONQUEST	No v4.0 member; the conditions are scenarios S-C1 to S-C6	No arbitration execution. Owned by #16. A conquest target list containing a sibling's own ASIN is a catalogue defect routed to #16 as a failed target, not an arbitration case.
STOP-LOSS: CONCEDE OR DEFER	STOP-LOSS: CONCEDE OR DEFER	No arbitration execution, and it does not close a finding. Recorded as a driver watch, because a stop-loss on the owner's root usually precedes the inventory or CVR movement that does move the composite. This was the live cycle-1 verdict on the Bamboo root while its ownership was undeclared.
PAUSED-PENDING (REASON + RE-ENTRY)	PAUSE (LEVERS EXHAUSTED)	No arbitration execution, and it never closes a finding. A paused campaign can be re-enabled, and the ownership question returns with it. The finding closes on a declaration or on the row below, not on a pause.

LEAVE TO HARVEST	LEAVE TO HARVEST	No arbitration execution, and the one verdict that closes a finding without a declaration. Where neither sibling will fund the root, the finding closes as declined-and-logged, the coverage state is written to both siblings' tab T2 block 1 "Coverage State" cells, and the guards lift because there is nothing left to guard.

The three v4.0-only strings this SOP issues

These are quoted from v4.0 Section 13.1 "The Closed Verdict Vocabulary" in place, with the source named, exactly as SOP-22 Section 5 and SOP-23 Section 5 both do under Ledger conflict C18. C18 registers two such strings; this session finds a third and reports it.

Verdict (v4.0 Section 13.1, verbatim)	KDR Template v3.0 member	Arbitration execution
PORTFOLIO CALL: ASSIGN LEAD	None (Ledger conflict C18)	The verdict this component exists to issue, at v4.0 scenarios S-B2, S-B3 and S-B5. P8 in full: the declaration record with its composite table, the three arithmetic re-arbitration triggers, the same-day wall instruction at P9, the non-owner posture at P10, and the tab T6 prediction. Human-mandatory per the Authority Matrix v2.0 row "Sibling declarations (S-series)".
CONTINUE; VERIFY MOMENTUM	None (Ledger conflict C18)	Issued at P14 where a live declaration's drivers held against their recorded triggers. The declaration re-dates with the refreshed composite table and the next review date; this is a real outcome and is recorded as one, not as an absence of action.
EXTEND TEST	None (this session's finding, same shape as C18)	Issued at P6 where the two composites have not separated, and at P5 where a probe tranche has not reached the sufficiency bar for a CVR read. The guards hold, no declaration files, and the re-read is dated. WE-2 is the worked case.

> **Three arbitration states carry no string in either closed vocabulary:** the stewardship grant, the guard-in-force state, and the re-arbitration outcome. SOP-33 v1.0 wrote local strings for all three plus two more; all five are retired at v2.0 as members of no closed vocabulary, following the precedent set when #23 retired three of its own. The gap is amendment item AM-10 and the states are recorded as gate outcomes on the finding until v4.1 resolves it.

6. Worked Examples

WE-1 and WE-4 are built on the anonymized live spine in the Economics and Context Pack v1.0 tab D1 and the Account Rollup Workbook v1.0 tab A1, with the sibling side disclosed as illustrative where that product has not yet run a cycle. WE-2 is built on the account's recorded sibling-divergence incident. WE-3 is illustrative on the standard structure. Every figure below recomputes from the four composite inputs stated beside it.

WE-1 · Clean case: the Bamboo root, declared

Setup. Root "bamboo sheets", search volume 101,819, primary syntax, purchase intent, oversight flag Y. Live in two family siblings' campaigns, so P1 condition 1 fires. The Family Conflict cell on the 4-piece workbook has read the open state with the 6-piece sibling named since cycle 1, and the "Term Owner (Family)" cell has read undeclared for the same period. Economics from tab D1: contribution $28.37 per order on AOV $81.05, so break-even ACOS = 28.37 / 81.05 = 35.0031%, Target = 17.5015%, Maximum Acceptable = 26.2523%. The 4-piece figures are live; the 6-piece figures are illustrative because that workbook has not run a cycle.

P4 inputs. 4-piece: margin $28.37 per order (blended row, so gate G9 rides PROVISIONAL onto this composite); root CVR 7 orders on 160 clicks = 4.375%; inventory depth 47 days on the KING-WHITE preferred SKU. 6-piece: margin $41.60 per order; root CVR 2.91%; inventory depth 39 days.

```
P6 composite (Composite = Margin x CVR x Depth; units relative)
  4-piece = 28.37 x 0.04375 x 47   = 58.3358125  ->  58.34
  6-piece = 41.60 x 0.0291  x 39   = 47.2118400  ->  47.21
  lead    = 58.3358125 - 47.2118400 = 11.1239725 -> 11.12, which is 23.56% of the 6-piece composite
  total   = 105.5476525
  share: 4-piece 55.27%   6-piece 44.73%
```

P7 inspection, all six steps. Step 1: both instances carry declared objective tags, so no untagged divergence. Step 2: the 4-piece top-of-search modifier sits at its evidence level and the 6-piece carries none, so effective top-of-search CPC is not the explanation. Step 3: both on the same bidding strategy. Step 4: duplicate flag N on both; no legacy instance bidding. Step 5: both point at their D2 preferred SKU. Step 6: reached and recorded, the residual difference is genuine auction position. The inspection clears and the composite stands.

P8 declaration. The 4-piece owns the root at 55.27% of composite. The three re-arbitration triggers solve the composite for each 6-piece input in turn against the owner's 58.3358125:

```
CVR trigger:    58.3358125 / (41.60 x 39)     = 3.5956%  ->  3.60%  (a 23.56% relative lift from 2.91%)
depth trigger:  58.3358125 / (41.60 x 0.0291)  = 48.18911 ->  48.19 days
margin trigger: 58.3358125 / (0.0291 x 39)     = $51.40172 -> $51.40 per order
```

Any one of the three re-opens the declaration under P12. All three are on the record as numbers, so the 6-piece owner can see exactly what would have to move, and so can the reviewer.

P9 and P10. Wall instruction issued the same session to #29 with the full source list on the 6-piece side: family broad, family phrase, both autos and the sibling discovery structure. Confirmation returns the same day and attaches to the record; only then do both tab T2 block 8 "Term Owner (Family)" cells update and both Family Conflict cells clear. The 6-piece keeps its count-modifier space and holds exact defence at floor there under DEFEND AT FLOOR, with those variants enumerated on the record at declaration time.

What the declaration does not fix. The owner's own root economics stay bad and stay visible: spend $522.05 on 160 clicks is a CPC of $3.26 against a maximum profitable CPC of 28.37 x 0.04375 = $1.24, which is 2.63 times the ceiling; 7 orders give a CPA of $74.58 against $28.37 of contribution; ACOS on the root is 522.05 / (7 x 81.05) = 92.02%, a ratio of 2.63 against break-even; organic rank is 39 against a target of 4 and the recorded cost to close that gap is $88,495. Ownership assigns the root; it does not authorise the spend. The owner carries BID DOWN TO PROFITABLE CPC; CAP SYNTAX on its own row in the same cycle, and the recorded cycle-1 verdict on this root was STOP-LOSS: CONCEDE OR DEFER. A reader who takes a declaration as permission has misread the component.

WE-2 · Edge case: the satin divergence, where the composite would have declared the wrong owner

Setup. The account's recorded sibling-divergence incident, carried in v4.0 Section 16 "Three Worked Passes: The Full System on the Account's Real Failure Patterns" as Worked Pass 2. Root "satin sheets queen" runs on two siblings. Product A: CPC $3.24, organic rank 17. Product B: CPC $1.48, organic rank 6. The CPC ratio is 3.24 / 1.48 = 2.19 times, above the v4.0 Section 4 rule X6 trigger and above the instrument's own divergence flag. Neither instance carries an objective tag on the term.

P4 inputs as first pulled. A: margin $34.10 per order from the product blended row; root CVR 21 orders on 402 clicks = 5.2239%; depth 58 days. B: margin $26.40 per order; root CVR 24 orders on 537 clicks = 4.4693%; depth 51 days.

```
P6 composite, pre-inspection
  A = 34.10 x 0.0522388 x 58 = 103.3179104 -> 103.32
  B = 26.40 x 0.0446927 x 51 =  60.1743017 ->  60.17
  A leads by 43.1436088 -> 43.14, which is 71.70% of B
  share: A 63.19%   B 36.81%
```

On that arithmetic the root goes to A, the product sitting at rank 17 and paying $3.24, and the walls fence B, the product sitting at rank 6 and paying $1.48, out of a root it already holds. P7 is what stops that.

P7 inspection, in the fixed order. Step 1 fires: neither instance is tagged, so the divergence has no stated reason and objective assignment is routed. Step 2 fires: A's top-of-search modifier reads +180% on a base bid of $1.75, so effective top-of-search CPC = 1.75 x 2.80 = $4.90. Step 3 fires: A runs up-and-down bidding, B runs fixed, so the two are not bidding the same auction. Step 4 fires: one legacy duplicate exact on A is still bidding, and its clicks and orders sit inside A's pulled CVR read. Step 5 fires and is decisive: A's advertised SKU is the parent, not the D2 preferred king variation the root actually lands on, whose contribution is $19.85 per order rather than the blended $34.10. Step 6 is never reached, so no auction explanation is admissible.

```
P6 re-run on the corrected margin input
  A = 19.85 x 0.0522388 x 58 = 60.1425373 -> 60.14
  B = 26.40 x 0.0446927 x 51 = 60.1743017 -> 60.17
  B leads by 0.0317644, which is 0.0528% of A
  share: A 49.99%   B 50.01%   ->  share gap 0.0264 percentage points

A's ceiling on the corrected margin
  max profitable CPC = 19.85 x 0.0522388 = $1.03694 -> $1.04
  effective TOS CPC $4.90 vs ceiling      = 4.73 times
  observed CPC     $3.24 vs ceiling       = 3.12 times
```

Outcome. The 71.70% lead collapses to a 0.0264-point dead heat, which is not a decision. The verdict is EXTEND TEST, quoted from v4.0 Section 13.1. No declaration files, the P3 guards hold on both siblings, and five repairs stage through their owners: objective assignment, the multiplier cut to evidence level, the bidding-strategy alignment, the duplicate consolidation at #12, and the re-point to the D2 preferred variation. The re-read is dated to the first cycle after all five land and a validated window accrues. The near-tie itself is logged as amendment item AM-1, because v4.0 states no minimum separation and its two stated tiebreaks do not agree with each other.

The lesson. The composite is only as good as the four numbers entering it, and the single most common defect is a margin read taken from the product blend rather than the variation the root lands on. A 42% error in one input produced a 71-point swing in a decision that binds two products' plans. Running P6 before P7 is not a shortcut; it is the mechanism by which settings archaeology becomes a governance record.

WE-3 · The probe: a sibling with no served evidence

Setup. Illustrative values on the standard structure. A Cooling family root is contested between a served 4-piece (product C) and a 6-piece (product D) that has never served the root, so D has no CVR on it and none may be imputed from C, from a modifier term, or from D's product average.

The ceiling test at P5 step 2. Before choosing between running a probe and declaring the served sibling with a dated re-check, compute D's arithmetically best possible composite using its own margin, its own depth and the highest CVR any family member has recorded on the root.

```
C composite  = 22.90 x (9/232) x 63 = 22.90 x 0.038793103 x 63 = 55.9668103 -> 55.97
D ceiling    = 31.20 x 0.038793103 x 52                        = 62.9379310 -> 62.94
62.94 > 55.97  ->  the ceiling clears the served sibling, so Path B is illegal and the probe is mandatory
```

The probe, run in two tranches. Bid reference is D's own Expected CPC, which reads at the account reference of $2.80 because the per-term value has not landed; the substitution is stated on the finding. Tranche 1 runs to the verdict string's cap of 50 clicks at $2.80 = $140.00 and returns 2 orders, a CVR of 4.00% on a sample the gate marks thin for a CVR read. Tranche 1 therefore cannot supply the composite input, and treating it as if it could is the F7 condition. Tranche 2 runs a further 50 clicks at $140.00, taking the cumulative sample to 100 clicks and 5 orders, a CVR of 5.00% that clears the sufficiency bar.

```
D composite after the probe = 31.20 x 0.05 x 52 = 81.1200000 -> 81.12
C composite                 = 55.9668103 -> 55.97
D leads by 25.1531897 -> 25.15, which is 44.94% of C
share: D 59.17%   C 40.83%

probe cost      = 100 x $2.80          = $280.00
contribution recovered = 5 x $31.20    = $156.00
net cost of the probe                  = $124.00
```

Outcome. D owns the root at 59.17% of composite, and the unserved sibling turns out to be the correct owner. Had the CVR been imputed from C at 3.8793%, D's composite would have read 62.94 and the answer would have been directionally right by luck; had it been imputed from D's product average, the answer could have gone either way. The probe cost $124.00 net to convert a guess into a record, against a declaration that binds a family root carrying real spend on both sides. This is why the composite never imputes a CVR.

WE-4 · Temporary stewardship, and the reversion that nearly did not happen

Setup. The declared owner on the "bamboo sheets full size" root is the 4-piece, and its serving variation is the FULL preferred SKU, which reads 18 days of cover at cycle with the inventory gate arming on 2026-07-10 at a projected 14 days and a projected stockout of 2026-07-24. The D2 map names a pre-pointed backup. Owner margin $28.37 per order, root CVR 4.12%. Steward candidate is the 6-piece at margin $41.60, root CVR 3.33%, depth 51 days.

P13 step 1, re-point first. The re-point to the named backup executes at #12 on the gate flip, which is an inventory move and not an ownership move. Stewardship is only reachable once the backup chain is exhausted, which it is on 2026-07-24.

```
P13 step 2, composites at grant
  owner (depth 18)  = 28.37 x 0.0412 x 18 = 21.0391920 -> 21.04
  steward (depth 51)= 41.60 x 0.0333 x 51 = 70.6492800 -> 70.65
  steward ahead by 49.6100880, which is 235.80% of the owner

P13 step 3, the reversion date
  restock confirmed          2026-08-14
  recovery period read from #24   21 days
  reversion date             2026-09-04
  grant duration 2026-07-24 to 2026-09-04 = 42 days
```

The record states plainly that the steward's 235.80% advantage at grant is an inventory artifact. Nothing about the steward improved; the owner's depth term collapsed from 47-class cover to 18 days, and depth is a multiplier. That sentence on the record is what keeps the grant reverting.

The near-miss at reversion. On 2026-09-04 the steward proposes to retain the root, citing its own performance across the 42 days. F5 fires: a stewardship past its reversion date is not an ownership case, and the reversion executes first. Re-running P12 on current inputs after the reversion:

```
owner restocked (depth 74) = 28.37 x 0.0412 x 74 = 86.4944560 -> 86.49
steward (unchanged)        = 41.60 x 0.0333 x 51 = 70.6492800 -> 70.65
owner reclaims by 15.8451760 -> 15.85, which is 22.43% of the steward
share: owner 55.04%   steward 44.96%

depth-only effect on identical margin and CVR: 86.4944560 / 21.0391920 = 4.11 times
```

Outcome. The owner holds the root on the legal route. The steward's case was never weak; the owner's case was temporarily absent, and the two are not the same thing. The 4.11-times swing on identical margin and CVR is the number to remember: depth alone moves the composite by more than any realistic margin or CVR movement will, which is exactly why a stockout must be handled as a dated grant rather than as a composite outcome. Walls move twice, both dated at grant per P9 step 5, so the reversion is staged rather than remembered.

7. Failure Modes and Escalation

Twelve modes. Each carries its detection signal, its response, and the account-history incident that motivated it where one exists. Six are carried forward from SOP-33 v1.0 and six are new at v2.0.

Failure	Detection signal	Response	Motivating incident
F1. Push open on a contested root	A live push objective on a root carrying an open finding, read on tab T2 block 7 against the finding register.	The push holds at the #23 precondition, not at this SOP. The finding fast-tracks to P6 with the composite prioritised over the queue. Escalates same day.	The named account pattern: the same term running at $3.24 on one sibling and $1.48 on another with no stated reason for either, recorded in v4.0 Section 1 as one of the failures the system was written to close.
F2. Declaration by history	A declaration record with no composite table attached, or with a table missing one of the four inputs on either side.	The declaration is void, not provisional. Guards re-arm, P4 re-runs, and the record is rewritten. Seniority, tenure and product size do not declare.	Carried from v1.0. The Bamboo root sat open across the first cycle with the declaration cell reading pending entry, precisely because nobody would write the table.
F3. Walls lagging the declaration	The declaration date and the wall state disagree; the #29 confirmation line has not returned by end of the declaration day.	Walls issue immediately. The declaration stays in drafted state, and both siblings' gate cells stay open until the confirmation attaches. The same-day rule is the entire mechanism.	Carried from v1.0, where it was recorded against the counterpart failure mode in #29's own table.
F4. Over-walling the non-owner	The non-owner's impressions collapse on its enumerated model-specific variants in the window after a wall build.	The P10 boundary is restored through #29's removal control with the declaration record as the evidence. Ownership is of the root only. The enumeration written at declaration time is what makes the restoration mechanical.	Carried from v1.0. A 6-piece set fenced out of its own count modifier because the family root moved to the 4-piece.
F5. Stewardship past its date	A live grant whose reversion date has passed, or a grant record with no reversion date on it at all.	Revert per the dated record, then run P12 if an ownership case genuinely exists. A grant with no reversion date is this condition at the moment it is written, not at the moment it lapses.	Carried from v1.0. Stewardship converting silently into ownership is how an inventory event becomes a permanent family decision nobody made.
F6. Stale composite	A named driver has moved past its recorded trigger with no P12 event logged, found at the P14 quarterly read.	Re-arbitrate. The quarterly review exists to catch what the triggers missed, and a review that finds drift without routing it has converted a control into a report.	Carried from v1.0.
F7. Composite on an imputed CVR	A composite row whose CVR cell has no click and order count behind it on the same sibling, or a probe tranche below the sufficiency bar used as a CVR input.	Void the composite. Run P5 properly in tranches. A CVR is never taken from the sibling, from a modifier term, or from the product average.	New at v2.0. WE-3 shows a case where imputation from the counterpart would have been directionally right by luck and imputation from the product average could have gone either way.
F8. Auction explanation accepted before the inspection ran	A declaration record whose P7 section is absent, partial, or records steps as clean without values beside them.	The declaration is void. P7 re-runs in the fixed order with a valued line per step. A step recorded as clean with no number is not a record.	New at v2.0, motivated by the recorded satin divergence, where five of the six inspection steps fired and the sixth was never reachable. WE-2 is the case.
F9. Finding open past its deadline	The finding register carries an open state with a deadline date in the past.	Escalate with the days-open counted, per P14 step 6. Guards stay armed; they do not lapse with the deadline. No v4.0 threshold sets the escalation clock, which is amendment item AM-12.	Carried from v1.0. Every undeclared week is the named failure pattern running live, funded by both siblings at once.
F10. Twins running split postures	Two terms sharing a twin-set identifier carrying different objectives, different bid postures, or simultaneous experiments across the family.	Consolidate to one posture per twin set before the composite reads CVR, because sufficiency pools across the set and a split posture makes the pooled read meaningless. Sequencing rule enforced at P11 step 4.	New at v2.0. The live workbook shows twin folds executed on cooling and bamboo roots in a single cycle, several of them with the reporting rolled under a parent after the fact rather than before.
F11. Provisional margin carried without its flag	A declaration record whose margin input traces to the blended row on the D1 table while the record carries no gate G9 flag.	Re-issue the record with the flag in the same words the engine computed. v4.0 Section 13.3 rule J7 makes hiding a fallback a process violation, and a declaration is the least appropriate place in the system to hide one.	New at v2.0. The live D1 table carries a blended row with a version stamp and variation rows that are still unfilled, so every current composite runs provisional and the flag is the only thing distinguishing that from a clean read.
F12. Scan not run, or run on a stale registry	No scan artifact for the cycle, or a scan artifact whose registry timestamp predates the cycle refresh.	Re-run against the refreshed registry. Record the tested-root count even when the finding count is zero, because a scan with no artifact is indistinguishable from a scan that never happened.	New at v2.0. The sibling view feed on the keyword tab reads as a pending build across the corpus, so the scan currently depends on the search-term join, and an unrecorded scan

hides which of the two sources actually ran.

Escalation. F1 and F5 escalate on first occurrence, because both are live spend running against a decision the system has already made or is about to make. F9 escalates on the deadline with the days-open counted. F2, F8 and F11 route to the PPC Lead with the void record attached, because all three void a governance artifact rather than a campaign setting. F4 routes to #29's wall audit with the declaration record as evidence. The remainder enter the next weekly pass worklist. Any second occurrence of the same mode on the same family inside one review period escalates regardless of class, because a family that repeats a mode is telling the account something about its structure rather than about its operator.

8. Outputs and Artifacts

Gates clear with artifacts, not assertions. Every row below names the record, where it is stored, how long it is kept, and who consumes it downstream.

Artifact	Storage location	Retention	Downstream consumer
Weekly scan record	Account Rollup Workbook v1.0 tab A1, scan log beneath the family grids.	Rolling four quarters, then archived with the register.	The PPC Lead at the P14 quarterly read, to confirm coverage rather than assume it; the F12 detection reads this row's absence.
S-series finding with evidence snapshot	Tab A1, in the instrument's own finding format.	Held open until closed by declaration or by declined-and-logged; retained one year after close.	#23 reads the open state as its push-block precondition; the P14 review reads deadlines and days-open.
Composite table	Tab A1 family grid row for the contested root, with the four inputs, their source cell references and the share column.	Every version retained; a re-arbitration appends rather than overwrites, so both tables sit side by side.	The declaration record; the Brand Management owner's review; the P12 comparison at re-arbitration.
P7 inspection record	Attached to the finding, one valued line per step, all six steps present.	Retained with the declaration for its full life.	#12 and #28 receive the named structural defects; the F8 detection reads this record's completeness.
Declaration record	Tab A1 declaration register, with the composite table, the three arithmetic triggers, the flags, and the wall confirmation attached.	Permanent. Superseded declarations are retained, not deleted, because the transfer history is the only evidence that ownership moved for a reason.	#29 for walls; #23 for push release; #12 for the gate G11 check; both products' workbooks for the owner cells.
Term owner and family conflict cells	Product Command Workbook tab T2, block 8 "Term Owner (Family)" and block 2 "Family Conflict", on both siblings.	Live state, refreshed each cycle; history lives in the decision log.	The gate G11 check #12 runs before any scale action on a shared term; the weekly pass reads them as gate columns before metrics.
Probe design and result	Attached to the finding; the campaign-level rows sit in the executing product's tab T2 block 7 with the test status and stop date.	Retained with the finding.	P4 as a qualified CVR input; #12 for the capped test execution.
Stewardship grant record	Tab A1 stewardship section, five fields including the computed reversion date and the restock ETA it derives from.	Retained two years, because a lapsed grant is the failure this record exists to make visible.	#29 for the dated wall moves in both directions; #24 for the recovery interlock; the daily reversion check.
Twin-set registry entries	Tab A1 twin-set registry, mirrored to tab T2 block 1 "Twin-Set ID".	Live state.	Sufficiency pooling at gate G2; the one-strategy rule at #12 and #29.
Decision-log entries with predictions	Product Command Workbook tab T6, append-only, with the rule or scenario identifier, the prior and new state, the expected outcome and the review date.	Append-only and permanent; the cycle file snapshot freezes it for the cycle.	V5 scoring at #6; the verification scoreboard on Account Rollup tab A3; the next cycle's opening step.
Quarterly register diff	Tab A1, as declarations added, changed, re-dated and unchanged with the composite share on each.	Retained four quarters.	The PPC Lead and the Brand Management owner; the amendment routing at P14 step 7.
Amendment register	Section 10.3 of this document, carried forward into each increment.	Until each item is resolved in a v4.0 release.	The v4.0 threshold owner, per the Authority Matrix row for threshold changes.

The gate artifact for a declaration is its record with the wall confirmation attached. The gate artifact for a cycle is the scan record. The gate artifact for a quarter is the register diff. A stewardship's gate artifact is its reversion, executed and logged.

9. Skill-Conversion Block

Block	Specification
Input bindings	Tab T2 block 1 "Twin-Set ID", "Syntax Tags", "Coverage State"; block 2 "Family Conflict", "Inventory Band", "Days to Stockout", "Targeted Variation(s)"; block 4 organic rank and target rank; block 5 cross-type clicks, orders and spend on the root; block 6 "Margin $/Unit", "Max Profitable CPC", "Expected CPC", "Cost to Close Gap $"; block 7 "Objective"; block 8 "Duplicate Flag", "TOS Modifier %", "Sibling CPC View (D6)", "Term Owner (Family)". Tab T3 objective tag, bidding strategy, placement multipliers, state. Tab T4 effective top-of-search CPC. Economics and Context Pack tab D1 Contribution $/unit and Version stamp; tab D2 preferred SKU and backup SKU. Account Rollup tab A1 family grid and declaration register; tab A2 the funding queue. Tab T7 manifest statuses for groups B5, B8, D2, A2, A3 and B3. An absent feed suppresses its dependent step per the manifest; the skill never improvises around a missing feed and never estimates a composite input.
Deterministic logic	The root normalisation from syntax tags. The two detection conditions at P1 step 4. The composite Margin x CVR x Depth as the instrument computes it, read from the instrument and never re-implemented. The share calculation. The three re-arbitration triggers, each solved by dividing the owner composite by the product of the non-owner's other two inputs. The P5 ceiling test. The P7 six-step order and its stop condition. The reversion date arithmetic at P13. All thresholds, bands, scenario calls and verdict strings are v4.0's; all factor definitions are the instrument's. The skill version-pins both.
Human checkpoints	Drawn from the PPC Authority Matrix v2.0 rows that name this component's decisions. From the row "Sibling declarations (S-series)": the PPC Lead decides every declaration, re-arbitration and stewardship grant; the incoming operator executes; the Brand Management owner reviews in the daily huddle, which is review and not veto, with disagreements routing up rather than sideways. From the row "Push funding (the M5 queue)": guard state is supplied by this SOP but funding release sits with the PPC Lead inside the envelope and the CEO above it. From the row "Structure creation (all SOPs)": wall issue and twin-set registration are executed against a PPC Lead decision. From the row "v4.0 threshold changes": every amendment item routes to the CEO as decider with the PPC Lead executing and reviewing. AI-executable with audit: the detection scan, the finding intake and evidence snapshot, composite input assembly with source references, the composite and share computation, the three trigger solves, the P7 inspection read, the reversion date arithmetic, drift detection at P14, and the register diff. Human-mandatory without exception: every declaration, every re-arbitration outcome, every stewardship grant, every probe standup, and the wall instruction that follows a declaration.
Cadence	Weekly detection scan inside the Monday cascade, after the rollup refresh. Same-session finding intake and guard arming. Same-day wall issue and confirmation on every declaration. Daily reversion check on every live stewardship grant. Quarterly declaration review with the register diff. Amendment routing at each quarterly close.
Alarm specification	F1 through F12 each carry a detection signal from Section 7, an evidence-line format naming the root, the two siblings, the signal value and the gate or rule reference, and a routing tier. Same day to the operator queue: F1, F3, F5, F9. To the PPC Lead with the void record attached: F2, F8, F11. To the #29 wall audit: F4. Into the weekly pass worklist: F6, F7, F10, F12.
Skill outputs	Drafted findings with evidence snapshots; drafted composite tables with source references and every flag carried; drafted declaration records with the three triggers solved and the non-owner variants enumerated; drafted wall instructions with the full source list for human confirmation; probe designs with tranche caps and stop conditions; stewardship grants with computed reversion dates; the quarterly register diff; drafted decision-log entries with dated predictions; and F1 to F12 alarms with evidence lines.
Refusal conditions	The skill declines to produce a declaration draft on any of five states: an unqualified composite input, a degraded packet, an incomplete P7 record, a probe tranche below the sufficiency bar, and a composite separation the operator has not accepted. In each case it returns the finding with the blocking state named rather than a best-effort declaration, because a drafted declaration is the artifact most likely to be confirmed without being read.

10. Version Governance

10.1 Review triggers specific to this component

- Any change to the Account Rollup Workbook A1 composite formula, its factor definitions, its grid columns or its finding format. The instrument owns the arithmetic, so an instrument change increments this document whether or not the process changes.
- Any v4.0 increment touching Section 5 gate G11, Section 8.11 sibling and portfolio scenarios, Section 10 rule M5, Section 12 precedence rule P9, or Section 13.1. The Section 5 verdict mapping re-validates against the new verdict set before any pass runs on it.
- Resolution of Ledger conflicts C10 or C18, either of which changes which strings this SOP may quote and from where.
- The landing of Ledger dependency D2, the keyword-to-variation map with backup chains, which lifts the depth input from a product-level substitution to the operational read and removes a stated cap on this component's depth.
- The landing of the per-variation margin table clearing gate G9, which removes the provisional flag currently riding on every composite.
- Any new sibling entering a family, or any family reorganisation, because both change the population the detection scan runs against.
- An increment to #29 or #23, since both carry an interface contract with this component on their own side and both have moved procedure numbers once already.
- Two re-arbitrations on one root inside a single review period, which triggers a structural review of the family rather than a third arbitration.

10.2 What changed at v2.0

Supersedes v1.0 in full. The composite method is corrected from a two-term value-per-click with a separate inventory factor to the three-term product the instrument computes, verified against the instrument's own live cell to seven decimal places; the v1.0 factor method reproduces neither the cell nor the doctrine statement. Composite units are corrected from dollars to a relative index. Procedures expand from 7 to 14, adding the detection scan as a procedure rather than a trigger, finding intake, input qualification, the probe with its ceiling test and tranche structure, the structural inspection as a mandatory pre-declaration step, wall issue and confirmation as a separate contract, the non-owner boundary, and the quarterly review with escalation. Failure modes expand from 6 to 12. Worked examples expand from 1 to 4, two of them on the anonymized live spine. The verdict mapping expands from 5 local strings to the full twenty-five-string closed vocabulary with the v4.0 cross-map column and three v4.0-only rows. Five v1.0 verdict strings are retired as members of no closed vocabulary: DECLARE OWNER, PROBE FIRST, RE-ARBITRATE, TEMPORARY STEWARDSHIP and GUARDS IN FORCE. Elements 11 and 12 are added. The v1.0 worked example is retired because its arithmetic implements the retired method. Twelve amendment items are logged rather than written as numbers. Zero local thresholds by design.

10.3 Amendment register: twelve items for v4.1

Each item is a number this SOP needed and v4.0 does not contain. None is written into the document.

ID	The gap	Statement and why it binds
AM-1	No minimum composite separation	Highest priority. v4.0 Section 8.11 scenario S-B2 assigns a lead by M5 value and tiebreaks by margin-dollar class and inventory depth; Section 10 rule M5 tiebreaks by margin-dollar class and earning-potential class. The two tiebreak lists do not agree, and neither states a separation below which the composite has not decided. WE-2 lands at a 0.0264-point share gap. Until this lands, a dead heat is handled as EXTEND TEST with the guards held.
AM-2	No declaration deadline interval	

		Every finding carries a deadline and no threshold sets it. The instrument's family-grid review cadence is currently the only anchor, and an instrument cadence is not a governed threshold.
AM-3	No stewardship reversion interval	The reversion is restock plus recovery, and the recovery period is read from #24's contract. v4.0 states no reversion interval for a stewardship grant as distinct from a post-restock recovery, so the two are currently the same number by borrowing rather than by rule.
AM-4	No inventory-depth floor for ownership eligibility	Depth enters the composite only as a multiplier, so a sibling with rich margin and a few days of cover can still win a root it cannot hold. Doctrine states that an owner who cannot sustain the pressure is an owner in name only, but no floor makes that operational.
AM-5	No composite recency window	v4.0 does not state over what window the root CVR input is measured. The instrument uses its cycle window and gate G2 carries 7, 30 and 60 day buckets for sufficiency, which are different questions.
AM-6	No qualified-traffic definition for the root CVR	Which campaign types count toward the root CVR input is unstated. Including auto and broad serving of the root produces a different number from exact-only, and the two siblings may not carry the same mix.
AM-7	Probe sizing has no matching verdict string	The closed vocabulary caps a deliberate test at 50 clicks; gate G2 requires a larger sample before a CVR verdict is legal. A probe sized to produce the composite's CVR input therefore has no single string, which is why P5 runs it in tranches.
AM-8	No minimum driver movement re-opening a declaration	The three arithmetic triggers this SOP computes are equality points. v4.0 states no tolerance around them, so a driver crossing a trigger by any amount currently re-opens a declaration, which will produce churn on roots whose inputs sit near equality.
AM-9	Review cadence stated at two intervals	The declaration review runs quarterly in this SOP's lineage while the instrument layer describes the term-owner assignment basis as reviewed monthly. v4.0 states neither, so the corpus carries two cadences for one control.
AM-10	Three arbitration states carry no verdict string	The stewardship grant, the guard-in-force state and the re-arbitration outcome have no member in either closed vocabulary. Same shape as Ledger conflict C18, which registers two such gaps; this adds three more and reports EXTEND TEST as a fourth string absent from the KDR twenty-five.
AM-11	Divergence trigger stated at two ratios	v4.0 Section 4 rule X6 and Section 8.11 scenario S-B1 both set the same-term sibling CPC divergence trigger at one ratio; the instrument layer's family-grid flag column sets it higher. v4.0 governs, so the instrument's flag currently fires later than the rule requires and the intermediate band goes undetected.
AM-12	No escalation clock on an overdue finding	P14 step 6 escalates a finding past its deadline with the days-open counted, but no threshold states at what count the escalation changes tier. Until it lands, escalation tier is a judgment made per occurrence.

11. Operator Ramp

Built from the PPC Authority Matrix v2.0 rows that name this component's decisions. Four rows apply directly: "Sibling declarations (S-series)", where the PPC Lead decides, the incoming operator executes and the Brand Management owner reviews; "Weekly verdicts (established US)", where the incoming operator executes inside an assigned scope and the detection scan rides the cascade; "Push funding (the M5 queue)", where this component supplies the guard state that the funding decision must respect; and "Structure creation (all SOPs)", under which wall issue and twin-set registration execute. The matrix vacancy rule applies throughout: an unfilled role's decision right reverts to the PPC Lead and its execution to the PPC Manager until filled, so the ramp below is written against roles that may temporarily be the same person.

Week	Competence demonstrated	Evidence the checkpoint holder accepts
Week 1 **Observation and artifacts**	Reads the instrument and the register without operating either. Locates every composite input at its source. States which of the four inputs is currently provisional across the account and why.	Walks the reviewer through one live declaration record and one open finding, naming for each input the exact tab, block and column it came from. Reproduces the instrument's live composite cell by hand to two decimal places. Lists the account's open findings with their deadlines and days-open, unprompted. Names the three components that consume this SOP's outputs and what each does with them.
Week 2 **Supervised execution**	Runs the full sequence on one contested root with the PPC Lead reviewing every step before it lands. Executes the detection scan unsupervised.	Produces a scan record with the tested-root count and the source that produced each finding. Assembles four qualified inputs per sibling with source references and every flag carried. Runs the six-step inspection with a valued line per step, including the clean ones. Computes the composite and the three arithmetic triggers, and defends the trigger arithmetic. Drafts the declaration record and the wall instruction with the full source list. The PPC Lead confirms before the declaration files and before the wall issues; the Brand Management owner reviews the composite table in the huddle.
Week 4 **Unsupervised execution, audited**	Operates the weekly scan, intake, guards, composite and inspection without step-level review. Declarations, re-arbitrations and stewardship grants remain human-mandatory at the PPC Lead, which is a permanent split and not a ramp stage.	Four consecutive scan records with no gap. One finding carried from detection to a filed declaration with the wall confirmation attached the same day. One correct refusal: a composite the operator declined to declare on, with the blocking state named. The PPC Lead audits the quarter's records rather than the steps: composite tables complete, inspections valued, flags carried, triggers solved as numbers, non-owner variants enumerated, predictions logged with review dates.

Live-defence questions

These test the reasoning behind the thresholds and the method, not recall of values. An operator who answers with a number and no mechanism has not passed.

1. The composite multiplies three terms rather than adding them or weighting them. What decision does multiplication make that a weighted sum would not, and what happens to the arbitration when one term approaches zero?
2. A sibling wins the composite by 42% and holds nine days of cover. The other holds ninety. Explain what the composite has already done with that fact, and say what you would still escalate.
3. The structural inspection runs before the declaration, not after it. Give the failure that ordering prevents, and explain why running it afterwards would not catch the same failure.
4. A stockout hands the sibling a composite advantage of more than two hundred percent. Explain why that is not an ownership case, and what specifically on the grant record stops it becoming one.
5. You are told to impute a CVR for an unserved sibling from its product average, because the probe costs money and the answer looks obvious. Give the argument against, using the probe's actual cost and what the declaration binds.
6. The two composites separate by 0.03%. State what you issue, why that is not indecision, and what has to be true before the question can be re-asked.
7. The non-owner keeps its model-specific variants. Draw the line between the root and the modifier space on a real family root, and say who restores the boundary when a wall crosses it.
8. A declared owner is running at 2.6 times its own ceiling on the root it owns. Explain why the declaration is still correct and what verdict the owner carries in the same cycle.

9. Your margin input traces to a blended row. State what rides onto the record, what would have to land to clear it, and why suppressing the flag would be worse than the blended figure itself.
10. A number you need is not in v4.0. Describe exactly what you do, and say why writing a defensible estimate would be the more expensive choice.

12. Interface Contracts

Eighteen contracts. Each names the sending owner, the receiving owner, the artifact, the timing and the consequence when it does not arrive. Counterpart contract identifiers on the #29 and #23 sides are cited from their Cross-Reference Ledger rows rather than from their documents, because both were rebuilt after the copies attached to this session and their procedure numbers have moved.

Upstream, into this component

ID	Sender	Artifact	Timing	Non-arrival consequence
I-1	Data Operations	Raw Pull Pack search-term and impression-share joins feeding the campaign child rows across all sibling workbooks.	Monday 06:00 land, 09:00 refresh complete.	The detection scan cannot run on its working source. Open findings hold their guards; no new finding opens; a newly contested root runs undetected for a full cycle with both siblings funding it. Recorded as F12 with the manifest gap named.
I-2	Finance, via Economics and Context Pack tab D1	Contribution $/unit and Version stamp per active variation.	Monthly true-up and on any cost or fee event.	The margin input falls back to the blended row and gate G9 rides PROVISIONAL onto the composite and the declaration. Declarations remain legal but every one carries the flag; suppressing it is F11.
I-3	Logistics and Catalog, via tab D2	Keyword-to-variation map with the preferred SKU and ranked backup chain per root.	Monthly or on assortment change.	Depth reads at product level rather than at the serving variation, and the composite is capped to a structural read that the finding states. This is Ledger dependency D2, currently OPEN, and it is the single largest cap on this component's depth.
I-4	Data Operations, via tab T2 block 2	Inventory band and days to stockout on the preferred SKU, per sibling.	Daily.	The depth term cannot be refreshed and no stewardship reversion date can be computed or checked. Live grants suspend rather than continue, because an unchecked grant is F5 waiting.
I-5	Data Operations, via Account Rollup tab A1	The family grid refreshed after all product cascades close, with the composite formula intact.	Mondays, after the product workbooks close.	The composite has no instrument. Findings hold at input assembly, guards stay armed, and no declaration issues. Computing the composite outside the instrument is prohibited: the instrument owns the arithmetic.
I-6	Data Operations, via tab T7	Manifest statuses for groups B5, B8, D2, A2, A3 and B3.	Weekly with the refresh.	The composite would run blind to its own gaps, which gate G8 forbids. Absent a manifest, every composite downgrades to a flag and the finding records that it could not see its own completeness.
I-7	#22 launch-stage PPC	The launch handoff with the newcomer's keyword set.	Within one cascade of the handoff.	A newly launched sibling bids family roots undetected until the next full scan, which is the highest-risk window in the component because a launch structure is built to serve broadly.
I-8	#32 deal and event window	The armed event plan naming any contested root on either sibling.	On arming, before the window opens.	A deal window can adjudicate ownership by accident: whichever sibling the event funds takes the root's traffic and the composite reads the result as evidence. The root is held out of the window when the plan does not name it.
I-9	#29 negation architecture	The same-day wall confirmation line, recorded as that component's I-9 in Ledger row 29.	Same day as the wall instruction.	The declaration stays in drafted state and both gate cells stay open. The non-owner keeps serving the root while the register reads it as decided, which is F3 and is the most convincing form of a false record.
I-10	#24 re-ranking and recovery	Confirmed restock date and recovery class for the owner's serving variation.	On restock confirmation and on any ETA change.	The stewardship reversion date cannot be computed at grant, which makes the grant F5 at the moment it is written. Grants are not issued without it.

Downstream, out of this component

ID	Receiver	Artifact	Timing	Non-arrival consequence
I-11	#23 ranking-push operations	The guard state and, on close, the ownership declaration. Recorded as that component's I-3 in Ledger row 23.	Guard state at intake; declaration at filing.	Per the counterpart row, an indefinite hold on both siblings. The push block is a precondition there, so its absence does not release the push; it freezes it without an owner to release it.
I-12	#29 negation architecture	The declaration, re-arbitration or stewardship grant with the full source list. Recorded as that component's I-8 in Ledger row 29.	Same session as the declaration; same day for the wall.	No wall is built, so the non-owner's discovery keeps serving the root and the declaration is cosmetic. On a re-arbitration, the outgoing owner keeps a fence it should have lost and the incoming owner is fenced out of what it now owns.
I-13	#12 exact campaign operations	The declared owner written into tab T2 block 8 "Term Owner (Family)" and the cleared or open state in block 2 "Family Conflict".	On declaration, after the wall confirmation attaches.	The gate G11 check that component runs before scaling a shared term has nothing to read, so it scales on an undeclared root. This is the mechanism by which the account's named divergence pattern re-forms after being closed once.
I-14	#12 and #29, via the twin-set registry	Twin-set identifiers mirrored to tab T2 block 1.	At registration and on any family change.	Twins split budget and pool no evidence, so sufficiency reads fail on each half and a probe case is manufactured that does not exist. Shared negatives also fail to propagate across the set.
I-15	The cross-product funding queue on Account Rollup tab A2	The contested-root exclusion list.	Same cascade as intake.	The queue funds a push the guard forbids, and the funding release then has to be unwound against a live campaign rather than against a queue row.
I-16	#6 V5 prediction scoring, via tab T6 and Account Rollup tab A3	Decision-log entries carrying the metric, direction, magnitude, horizon and review date for every declaration, transfer and grant.	Same session as the action.	Unscored actions do not accumulate under v4.0 Section 14 rule V1. The component keeps declaring and never learns whether its declarations moved family blended CPC or family rank capture, which is the only evidence that the composite is calibrated.
I-17	The Brand Management owner	The composite table and declaration record for the daily-huddle review, per the Authority Matrix row "Sibling declarations (S-series)".	Before the declaration files.	The review happens without the arithmetic in front of it, which is F2 arriving through the review rather than through the record. The matrix makes this review, not veto: disagreement routes up with evidence, never into parallel spend.
I-18	The v4.0 threshold owner, per the Authority	The Section 10.3 amendment register, refreshed at each quarterly close.	Quarterly.	The register accretes without resolution and each session re-discovers the same twelve ungoverned numbers. The failure is silent: nothing

Matrix row "v4.0 threshold changes"	breaks, and the component simply keeps operating with twelve decisions that no threshold governs.

Appendix A. The Composite Worksheet Card

One card per contested root. The SOP text governs; this is the field order for the A1 grid row.

```
ROOT ______________________  FAMILY ______________  FINDING OPENED ____________  DEADLINE ____________

                           SIBLING 1              SIBLING 2
  Margin $ / order         __________  src ____   __________  src ____   [G9 flag? Y/N]
  Root CVR (orders/clicks) ____ / ____ = ____%    ____ / ____ = ____%    [sample state ____]
  Inventory depth (days)   __________  SKU ____   __________  SKU ____   [D2 mapped? Y/N]
  Window                   ______________________ (same both sides: Y/N)

  COMPOSITE = Margin x CVR x Depth
                           __________             __________             (units relative)
  SHARE OF COMPOSITE       _______%               _______%               separation _______%

  P7 INSPECTION  1 tags ____  2 eff TOS CPC ____  3 strategy ____
                 4 duplicates ____  5 variation ____  6 auction ____   (value on every line)

  RE-ARBITRATION TRIGGERS for the non-owner, each solved against the owner composite:
     CVR    = owner composite / (non-owner margin x non-owner depth)   = ______%
     depth  = owner composite / (non-owner margin x non-owner CVR)     = ______ days
     margin = owner composite / (non-owner CVR x non-owner depth)      = $______

  DECLARED OWNER ____________  DATE ________  WALL CONFIRMED ________  T6 PREDICTION ________
  NON-OWNER PROTECTED VARIANTS (enumerated at declaration): _________________________________
```

Appendix B. Weekly Operator Card

```
 1. Wait for the Account Rollup refresh. A scan before it reads last week's grid.
 2. Build candidates from both sources: the family columns AND the search-term join.
 3. Normalise to roots using the syntax tags. Modifiers are not the root.
 4. Two conditions, either alone opens a finding: live in 2+ siblings, or converting in 2+.
 5. Exclude on evidence only. Three legal exclusions, no fourth.
 6. Record the scan even at zero findings, with the tested-root count.
 7. On a finding: snapshot both siblings BEFORE arming guards, then arm the same session.
 8. Guards freeze escalation, not spend. They lift on a filed declaration only.
 9. Qualify all four inputs on both sides before computing anything.
10. Never impute a CVR. Run the ceiling test, then the probe in tranches.
11. Composite is the instrument's. Do not overwrite the formula cell. Units are relative.
12. Inspection BEFORE declaration, six steps in order, a value on every line.
13. Close separation is not a decision. Hold the guards and date the re-read.
14. Solve the three triggers as numbers. A direction cannot fire.
15. Walls the same day, with the FULL source list. Gate cells clear only on confirmation.
16. Enumerate the non-owner's variants at declaration, not after a wall suppresses them.
17. Every declaration, transfer and grant logs a dated prediction in the decision log.
18. A number v4.0 does not carry goes to the amendment register, never into the document.
```

Companions: Master PPC Decision Criteria System v4.0 (all thresholds, scenarios and verdicts) · Account Rollup Workbook v1.0 tab A1 (the composite instrument) · Keyword Decision Record Template v3.0 (the closed verdict vocabulary) · PPC Authority Matrix v2.0 (Elements 9 and 11) · Master PPC Data Architecture v1.0 and the PPC Command Workbook TEMPLATE v1.0 (instrument mechanics and artifact names) · Economics and Context Pack v1.0 tabs D1 and D2 (economics and variation map) · PPC Strategic Doctrine v1.0 Section 5.2 case 4 and Section 14 (the why) · PPC Cross-Reference Ledger (boundaries, and the source for every seam cited here) · SOP-29, SOP-23, SOP-12, SOP-24, SOP-32, SOP-22 (adjacent components, cited by Ledger row).